

COVID-19 Global Impact: Executive Summary

Veda Technology | Data Analyst Internship Project | Prepared by Trisha | 160 countries, Jan 2020 - Jun 2023 | Source: Our World in Data

Question: did faster vaccination rollout go with lower death rates?

Partly, but not on its own. Raw comparisons point the wrong way because older countries both vaccinated more and had more people at risk. Holding age constant, coverage shows no overall link to 2022 deaths, and only the oldest third of countries show fewer deaths with faster rollout. The clearer signal: deaths per reported case fell 71% for the typical country by the Omicron era, and fell more where coverage was higher.

Key insights

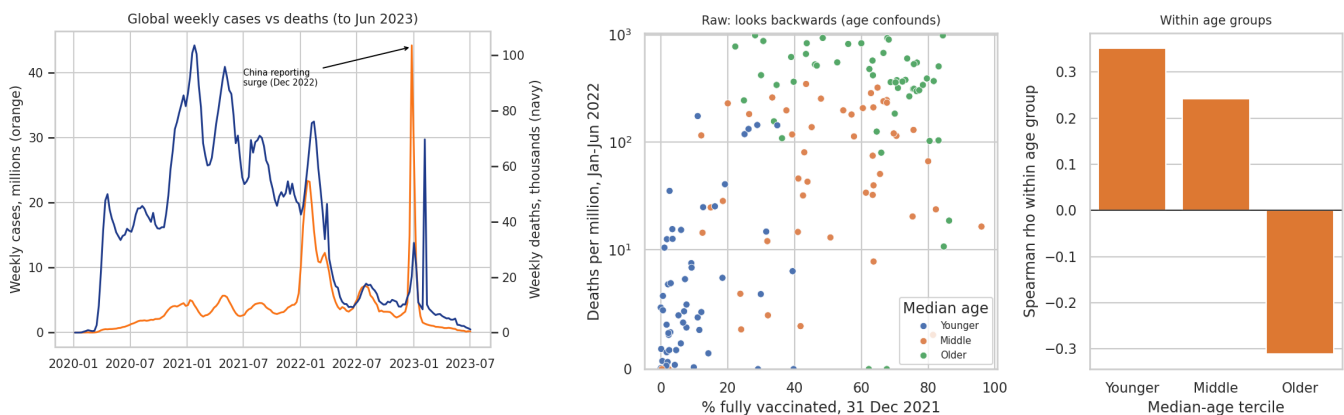
1. Deaths peaked a year before the Omicron case wave. Weekly deaths across the 160 countries peaked in the week of 24 Jan 2021 (about 104K/week). The Omicron wave a year later pushed cases to about 23M/week with far fewer deaths: deaths per 100 reported cases fell from 2.19 (original/Alpha) to 0.40 (Omicron) and 0.30 afterwards.

2. Reported burden was highest in the Americas and Europe, but reporting quality differs. By Jun 2023, reported deaths per million were 3,108 in South America, 2,779 in Europe and 2,679 in North America, versus 346 in Asia and 181 in Africa. Peru was highest at 6,476. Low figures in Africa and Asia partly reflect limited testing and death registration.

3. Raw data makes vaccination look harmful, but age explains it. Across 155 countries, higher vaccine coverage at end-2021 correlated with MORE deaths in Jan-Jun 2022 (Spearman $\rho +0.60$), because older, richer countries vaccinated more and also had more at-risk people. After adjusting for median age the link disappears ($\rho +0.00$, $p=0.95$).

4. Faster rollout lined up with fewer deaths only in the oldest countries. Splitting countries into thirds by median age: in the oldest third, higher coverage went with fewer deaths ($\rho -0.31$, $p=0.025$). In the youngest third the sign flips ($\rho +0.35$), likely reflecting differences in testing and reporting, not vaccines. Answer to the focused question: partly, and not on its own.

5. Deaths per reported case fell sharply, more so where vaccination was higher. The typical country's death rate per reported case in the Omicron era was 29% of its original-era rate (a 71% drop); it fell in 93% of countries. The drop was larger where end-2021 coverage was higher ($\rho -0.40$, $p<0.001$, $n=151$), though immunity from infection, milder variants and wider testing also play a part.



Method and caveats

Cleaned the OWID file (aggregates and countries under 1M people removed, duplicates dropped, cumulative gaps filled and revisions flattened), then rebuilt daily counts, 7-day averages, per-million rates, fatality rate and doubling time. Analysis stops at 30 Jun 2023 because many countries stopped reporting. Findings are associations, not proof of cause. Fatality rates use reported cases and depend on testing; low figures in Africa and Asia partly reflect under-reporting.